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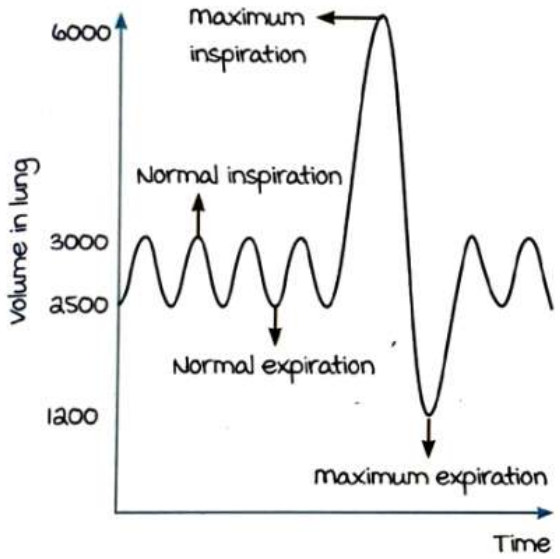
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# LUNG VOLUMES AND CAPACITIES

| | Time independent | Time dependent |
|---|---|---|
| AKA | Static volumes | Dynamic volumes |
| Eg | Tidal volume | $FEV_1$ (Forced expiratory volume in 1st second) |

## Static Lung Volume & Capacities

00:03:30




1. Inspiratory reserve volume (IRV) 2-3 L
2. Tidal volume (TV) 500 ml
3. Expiratory reserve volume (ERV) 1.3 L
4. Residual volume (RV) 1.2 L

Inspiratory capacity (IC) IRV + TV

Functional residual capacity (FRC)/ Equilibrium volume = TLC/2 or ERV + RV (2.5 L)

Vital capacity IRV + TV + ERV

Total lung capacity IRV + TV + ERV + RV (5 L)

5. Closing volume : Volume enclosed in lower dependent alveoli during the end of expiration.
   Closing capacity : Closing volume + residual volume.

**Investigations :**

1. Spirometry : Can measure all lung volumes except residual volume.
   Hence cannot calculate FRC & TLC.



2. Body plethysmography :
   Use : To measure RV.
   Principle : Boyle's law $P \propto 1/V$



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Interpretation :

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| Inspiration | Volume (V) | Pressure (P) |
|---|---|---|
| Lungs | ↑ | ↓ |
| Chamber | ↓ | ↑ |

Formula for calculating plethysmography :

$P_m i \times V_L i = P_m f \times (V_L i + \triangle V)$

$P_m i$ : Initial pressure in mouth = Atmospheric pressure = 760 mm Hg.

$V_L i$ : Initial volume in lung = FRC.

$P_m f$ : Final pressure at mouth.

$\triangle V$ : Change in volume.

Eg : If inspiratory effort lowers pressure at mouth by 10 mmHg & expands gas in lung by 50 ml, Calculate FRC ?

Ans : FRC = ?

$P_m i$ = 760 mmHg.

$V_L i$ = FRC.

$P_m f$ = 760 − 10 = 750 mmHg.

$\triangle V$ = 50 ml.

$P_m i \times V_L i = P_m f (V_L i + \triangle V)$.

$760 \times FRC = 750 (FRC + 50)$.

$760\ FRC = 750\ FRC + 37500$.

$10\ FRC = 37500$.

FRC = 3750 ml.

Note : Obsolete methods for calculating RV :

- Helium dilution technique.
- Nitrogen washout technique.

## Dynamic Lung Volume & Capacities 00:24:45

1. Forced vital capacity (FVC) estimation :

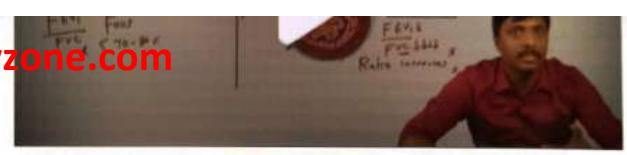
Deep inspiration —Followed by→ Fast & forceful expiration in 6 sec.

2. Forced expiratory volume in first second (FEV) :

- 70 − 80 % of air expired in first second in healthy individuals.
- Tiffeneau − Pinelli index : $\frac{FEV_1}{FVC} = 70 - 80\ \%$
  - Use : To differentiate obstructive (Difficulty in expiration) & restrictive (Difficulty in inspiration) lung diseases.

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Obstructive

$\frac{FEV_1 \downarrow}{FVC}$ = Ratio ↓ ↓



Early stage

$\frac{FEV_1 \downarrow}{FVC \downarrow}$ = Ratio normal



Late stage

$\frac{FEV_1 \downarrow}{FVC \downarrow \downarrow}$ = Ratio ↑ ↑

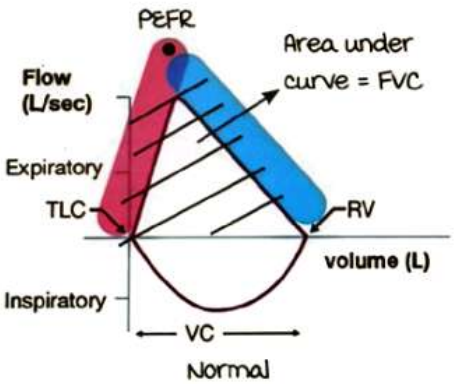
Restrictive

Note : Gold's criteria used for COPD is based on $FEV_1/FVC$.

## Flow Volume Loop

00:32:44

Normal lung :




Normal

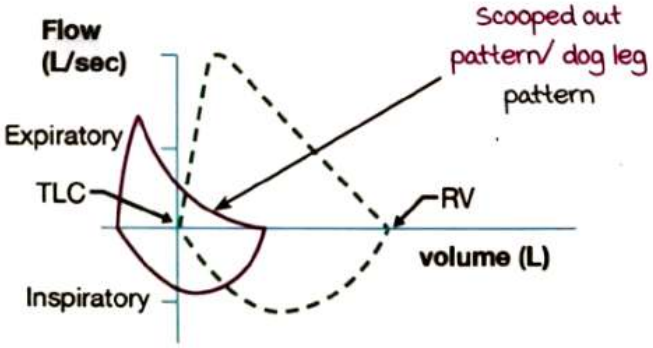
- Trachea
- Bronchi (medium & small size airway : most sensitive to obstruction)

FVC = Forced vital capacity

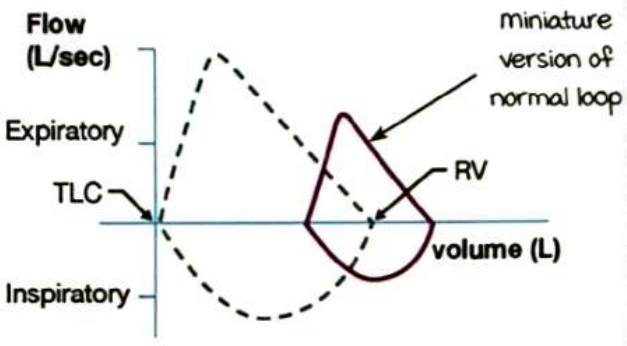
TLC - RV.

PEFR : Peak expiratory flow rate.

Obstructive vs Restrictive lung diseases :




Obstructive lung disease




Restrictive lung disease

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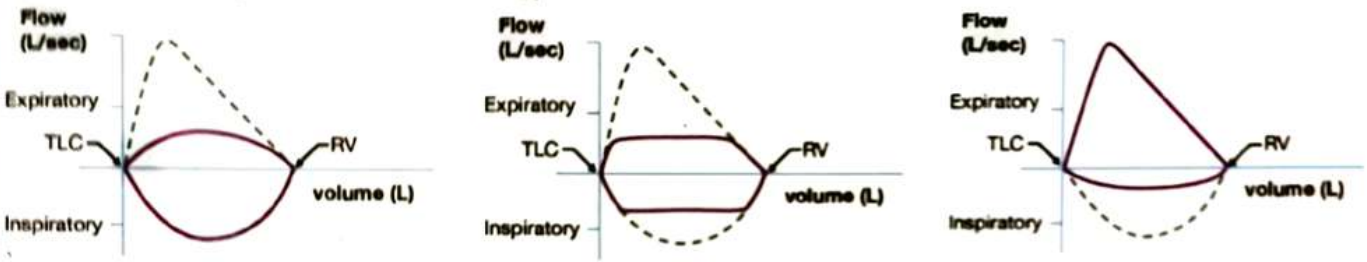


Airway obstruction :

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Intrathoracic (Foreign body obstruction)

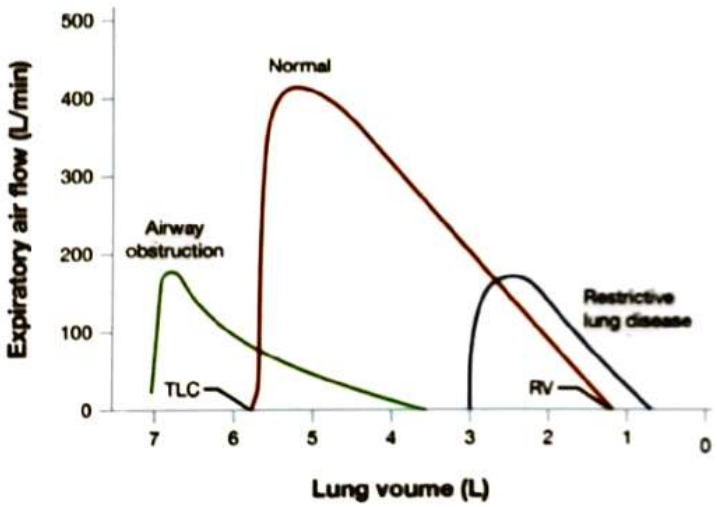
Extrathoracic (Sleep apnoea)




Variable intrathoracic LAO

Fixed upper airway obstruction (uAO)

Variable extrathoracic (uAO)




**TLC** - Total lung capacity
**RV** - Residual volume

Expiratory flow volume curve for airway obstruction vs Restrictive lung disease

Feedback